

HARINANDAN A

Mechanical Engineering Student · Piping & Stress Analysis · IoT & AI Projects

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PROFESSIONAL SUMMARY

Motivated Mechanical Engineering student (CGPA: 8.0) with hands-on expertise in piping stress analysis, FEA simulations, and engineering design, complemented by strong programming, IoT, and AI project experience. Completed the TATA Ready Engineer Programme and holds NPTEL certifications in Design Innovation and Lean Manufacturing. Known for analytical thinking, adaptability, and cross-disciplinary problem-solving.

TECHNICAL SKILLS · EDUCATION

Engineering & Design: SolidWorks, Caesar II, ANSYS, Engineering Drawing & Isometrics, API Codes 520 & 526

Programming & Tech: C, Python, Arduino UNO, IoT Sensors & Modules, Drone Technology

Standards: API 520 & 526, P&ID Interpretation, Piping Material Selection

BE Mechanical Engineering
[RV College of Engineering, Bangalore](#)
2023 – 2027 · CGPA: 8.0 / 10

Senior Secondary (CBSE)
[Kendriya Vidyalaya No.1, Calicut](#)
2022 · Aggregate: 89%

Matriculation (CBSE)
[Bharatiya Vidya Bhavan School, Calicut](#)
2020 · Aggregate: 95%

INTERNSHIP EXPERIENCE

Mechanical Engineering Intern — Larsen & Toubro Limited (L&T)

[Piping – E&M Department, EDRC](#) | [Heavy Civil Infrastructure](#) | [Chennai HQ](#) Aug – Sep 2025

- ▶ Performed stress and failure analysis using Caesar II and ANSYS, applying Von Mises criterion and API codes 520 & 526.
- ▶ Interpreted P&IDs and isometric drawings; studied valve classifications and piping material selection (CS, SS, GRP).
- ▶ Applied fluid mechanics, thermal expansion, and seismic load principles to real-world piping design scenarios.

FEATURED PROJECT

Flood Disaster Response & Rescue Priority System / Interdisciplinary Project · RVCE 2026

Stack: [Python](#) · [LSTM](#) · [XGBoost](#) · [IMD/CWC Data](#) · [Geospatial Mapping](#) · [ANSYS Fluent](#)

- ▶ AI-driven system for real-time flood rescue prioritisation using LSTM and XGBoost models trained on IMD/CWC data.
- ▶ Planned ANSYS Fluent simulation of flood flow dynamics to assess rescue asset feasibility as Phase 3 deliverable.
- ▶ Integrated mechanical and computer science disciplines in a four-person interdisciplinary team (2 ME + 2 CS).

PROJECTS

AI-Based Personalised Learning Platform using LLM RVCE · 2026

Built an AI-powered platform using LLMs for adaptive course content generation and intelligent student assessment.

IoT-based Smart Campus System RVCE · 2025

Occupancy tracking and automated environmental controls with a real-time web dashboard for energy optimisation.

Rotating Solar Panel with Microcontroller RVCE · 2025

Automatic solar tracking system using light sensors and Arduino UNO for maximised energy capture.

IoT-based Smart Inventory Management System RVCE · 2025

Warehouse tracking solution using IoT sensors for real-time inventory monitoring.

Pneumatic Power Steering System *RVCE · 2024*
Designed and tested a cost-effective pneumatic steering mechanism applying fluid power principles.

CERTIFICATIONS & TRAINING

- ▶ **TATA Ready Engineer Programme**
Fundamentals & Advanced Automobile Design
- ▶ **NPTEL Certifications**
Design Technology & Innovation | Lean Manufacturing
- ▶ **Skill Lab**
ANSYS (FEA & Simulation) | Best Performer – Sheet Metal & Welding

Languages
English, Malayalam, Hindi — Read / Write / Speak
Tamil — Speak

Extracurriculars
Astra Robotics Club (Technical Member) | **Rotaract Club** (Volunteer)
E-Cell RVCE | RAAG RVCE
Interests: Piping Systems · IoT Automation · AI/ML Research · Drone Technology